

Lab 2: Stereo, 5.1 and VBAP panning

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Introduction

In this lab session will cover 2.0 stereo and 5.1 stereo panning and compare it to VBAP panning. The core concept of this lab is the panning of virtual sound sources around the listener by means of psycho-acoustic principles. *Panning* techniques rely on the distribution of a sound signal over several loudspeakers to create the impression that the sound source is placed in a location where no loudspeakers are present (we refer to this kind of sources as "virtual" sound sources). The perceived location of the virtual source when using a specific loudspeaker layout follows known psycho-acoustic rules, from which we can derive panning techniques.

In this first lab session we will practice panning over two basic formats of multichannel audio: two-channel stereo (the standard format for FM radio, TV channels and Internet distribution) and 5.1 stereo surround. *Two-channel stereo* (or *2.0 stereo*, or simply *stereo*) has two frontal channels, *left* (L) and *right* (R), located at $\pm 30^\circ$ in its standard formulation (see Figure 1). In a 2.0 stereo audio file, the channels are in the order L-R.

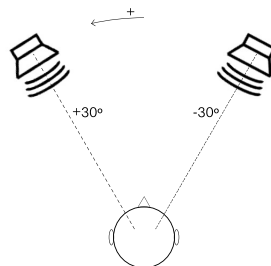


Figure 1: Standard stereo pair layout. ©

5.1 stereo surround (or *5.1 surround*, or simply *5.1*) adds four additional channels: a *centre* channel (C), to be located in front of the audience; *left surround* (Ls) and *right surround* (Rs) channels, to be located at $\pm 110^\circ$ approximately, and a *low frequency effects* channel (LFE), to be reproduced through a subwoofer. There are different standards for the order of the six channels in a multichannel 5.1 surround audio file; for this lab session we will use the channel order L-R-C-LFE-Ls-Rs.

Panning theory summary

The panning techniques that we will cover in this lab session are *amplitude panning* and *time panning*. Both theories assume that the listener is placed in front of the loudspeaker pair, in a position where the angular distance to each loudspeaker is equal (as in Figure 1).

Stereo amplitude panning relies in our perception of the location of a virtual source when comparing the same signal delivered by two loudspeakers simultaneously at different levels (Figure 2a). The virtual source is perceived in the arch between the two loudspeakers. The curve that expresses the relation between the level difference and the perceived azimuth angle can be modeled with the so called "Tangent law":

$$\tan(\theta) = \frac{(g_1 - g_2)}{(g_1 + g_2)} \tan(\theta_t),$$

where θ is the perceived angular position of the virtual source (in counterclockwise sense), θ_t is the angular aperture of the loudspeaker pair (e.g. 30° in a standard stereo pair), g_1 and g_2 are the normalised positive gains of each loudspeaker (e.g. $g_1 = g_{Left}$ and $g_2 = g_{Right}$ in a standard stereo pair).

Assuming a coherent sum, if we add the condition of a constant amplitude $g_1 + g_2 = k$, (with the normalisation condition $k = 1$), the system of equations has a unique solution in case of knowing two of its variables (e.g. when we want to know the gain values from known θ and θ_t).

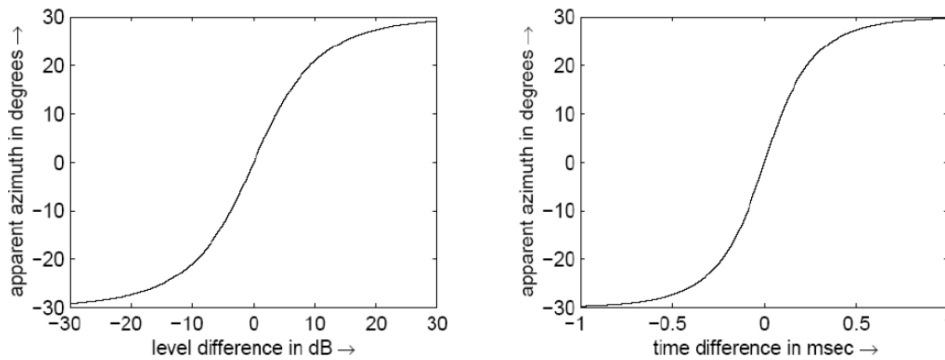


Figure 2: a) The left curve shows the psychoacoustical relation between the level difference between channels (in dB) and the perceived azimuth of the virtual source (in degrees) in a standard 2.0 stereo setup. b) The right curve shows the psychoacoustical relation between the time difference between channels (in milliseconds) and the perceived azimuth of the virtual source (in degrees) in a standard 2.0 stereo setup.

Stereo time panning is likewise based on the relationship between the time delay applied on one of the loudspeakers and the perceived location of the virtual source (Figure 2b). You can use the model shown in Figure 3. The formula for which is:

$$ITD = \frac{r(\theta + \sin(\theta))}{c}$$

Vector Base Amplitude Panning (VBAP) is a method for positioning virtual sources to arbitrary directions using a setup of multiple loudspeakers. In VBAP the number of loudspeakers can be arbitrary, and they can be positioned in an arbitrary 2D or 3D setup. VBAP uses at any one time the minimum number of

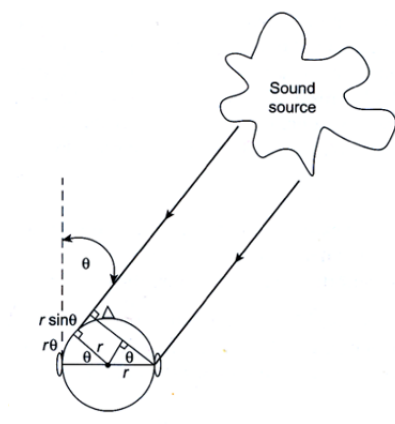


Figure 3: An approximation for the ITD using a spherical head model.

loudspeakers needed, be that one, two, or three. Its panning methods are equivalent to the Tangent law. However, its vector base approach to derive the gain factors gives it the flexibility to expand panning into the vertical dimension.

Description of the assignment

The work you have to do in this lab session is divided in three parts: a practical part, where we will collect knowledge and experiences by solving the panning problem together (and you should take notes for later), a programming part, in which you are asked to solve some scripts, and a written part, which you are asked to answer some questions related to the activities of the lab. You can finish the programming and written part at home, if necessary, based on the notes you took during the lab session itself.

Instructions

1. Attend to the lab session, listen carefully to the different audio samples, and, according to your own perceptions, take notes and answer the questions of this document (use only the space provided).
2. Save this document as `CODE_lab2.pdf` and upload it to *Aula global* before the deadline for part 2.
3. Choose a 5s mono segment of any audio file you like (if it is not mono, convert it to mono) sampled at 48 kHz. Save this mono segment as a wave file named `CODE_mono.wav` (substitute "CODE" by your actual group code).
4. Solve the tasks in following subsections, by writing simple audio manipulation scripts written in *Python*, starting with the mono file you have just created. It is strongly suggested to you test the results in any stereo reproduction system or headphones.
5. Fill in the appropriate answers to the questions in this section (use only the space provided). Save this document as `CODE_lab2.pdf`.
6. Archive all the audio files (remember to use a sampling rate of 48 kHz please), the source code files and this document in a zip file named `CODE_lab2.zip`. Upload this zip file to the *Aula Global* before the deadline.

Coding Stereo and 5.1 Panning

Amplitude panning in two-channel stereo

1. By using amplitude panning, generate a stereo file (of 20s duration, approximately) in which the virtual sound source is consecutively localized at:
 - a) 0° (this is, the front).
 - b) 10° on the right.
 - c) 20° on the right.
 - d) 30° on the right (this is, the right speaker).

Make sure that the perceived audio level is similar in each one of the cases, and similar to the original mono file.

2. Save your source code as `CODE_stereo_intensity.py` and the generated stereo file as `CODE_stereo_intensity.wav`.
3. Write down the gains applied to the loudspeakers in each case:
 - a) At 0° : $g_{Left} = 0.5$, $g_{Right} = 0.5$.
 - b) At 10° : $g_{Left} = 0.3473$, $g_{Right} = 0.6527$.
 - c) At 20° : $g_{Left} = 0.1848$, $g_{Right} = 0.8152$.
 - d) At 30° : $g_{Left} = 0$, $g_{Right} = 1$.

Time panning in two-channel stereo

1. By using time panning, generate a stereo file (of 20s duration, approximately) in which the virtual sound source is consecutively localized at:
 - a) 0° (this is, the front)
 - b) 10° on the left
 - c) 20° on the left
 - d) 30° on the left (this is, the left speaker)
2. Save your source code as `CODE_stereo_time.py` and the generated stereo file as `CODE_stereo_time.wav`
3. Which is the difference in the onset times of the two channels? (in samples and ms)
 - a) $t_L - t_R = 0$ samples, 0 ms.
 - b) $t_L - t_R = 4$ samples, 0.0833 ms.
 - c) $t_L - t_R = 8$ samples, 0.1667 ms.
 - d) $t_L - t_R = 13$ samples, 0.2708 ms.

Sampling rate used: `48000` Hz.

Intensity panning in 5.1 surround

1. By using intensity panning, generate a 5.1 multichannel file (of 20s duration, approximately) in which the virtual sound source is consecutively localized at:
 - a) 0° (front), using loudspeaker(s) **C** and **C**.
 - b) 60° on the left, using loudspeaker(s) **L** and **LS**.
 - c) 90° on the left, using loudspeaker(s) **L** and **LS**.
 - d) 180° (rear), using loudspeaker(s) **LS** and **RS**.

Hint: Use always the two loudspeakers next to the virtual sound source, except if the virtual sound source coincides with the location of a real loudspeaker (use only that loudspeaker in this case).
2. Save your code as `CODE_51_intensity.py` and the generated sound file as `CODE_51_intensity.wav`.

Questions on Stereo and 5.1 Panning

1. According to your perception, in the case of intensity panning, is the virtual sound source correctly localized from each one of the four directions? Is there any difference between the four cases? Why?

Generally yes, it works well in all four cases. The clearest positions are 0° (equal gains, stable centre) and 30° (single speaker, no ambiguity). The intermediate angles (10° , 20°) work but the phantom image is more fragile, moving your head slightly breaks the illusion. You must be in the sweet spot for it to work.

2. Same question for time panning: Is the virtual sound source correctly localized from each one of the four directions? Is there any difference between the four cases? Why?

Time panning works but feels weak since delays are tiny (max 13 samples at 30°), so the image shifts slightly left but stays inside your head. All three left angles sound similar. However, in order to achieve full externalized localization needs both time and level differences together.

3. Comparing time panning and intensity panning, which differences you perceive? Do you feel that any of the two is more appropriate than the other?

Intensity panning is clearly stronger and more convincing. Time panning alone produces a vague internalized shift, not a real sense of direction. Intensity panning wins for practical use. Nevertheless, in real spatial audio both cues are combined using ITD for fine localization and ILD for strength.

4. In the case of a frontally localized sound (0°), compare stereo and 5.1. Which advantages might 5.1 have over stereo for frontally localized sound?

In stereo, the centre image is a phantom that only works in the sweet spot. In 5.1 there is a real centre speaker, so the front image is stable for every listener regardless of position. This is why cinema always routes dialogue to the C channel.

Comparing VBAP

1. 2D panning. Listen in 2D, and locate a source at each of the following positions: 0, 45, 90, 110 and 180 degree (in the horizontal plane). Compare between VBAP and 5.1 panning. What differences do you hear? For VBAP, how many channels are active in each case? Why?

2. 3D panning. Listen in 3D using VBAP. Locate the source at different positions of your choice. What differences do you hear? How many channels are active in each case? Why?

3. Now, lets move the loudspeakers to arbitrary positions that are outside the norm. We recalibrate VBAP to accomodate for these positions. What differences do you hear when compared to 5.1 panning?

Questions on VBAP

1. "VBAP is a layout-independent paradigm". What does this sentence mean?

VBAP doesn't assume a fixed speaker layout like stereo or 5.1. You feed it the actual positions of your speakers, whatever they are, and it calculates the correct gains automatically. Therefore, the same virtual source position will sound correct regardless of how many speakers you have or where they are placed.

2. "VBAP is an object-based method". What does this sentence mean?

Instead of encoding audio as fixed channel signals, VBAP encodes sound as an object with a position in space. The gains for each speaker are calculated at render time from that position, in other words, the same audio object can be reproduced correctly on any speaker layout (layout independent) and thus you're not locked to a specific channel configuration.

3. What are the advantages of VBAP versus channel-based (e.g. 5.1) reproduction?

Unlike 5.1, where the mix is hardcoded for specific speaker positions, VBAP computes speaker gains from a source position. This means the same mix works on any speaker layout, improves automatically with more speakers, and extends to 3D space, things channel-based systems simply cannot do.

Implementation of a VBAP function in 2D

Complete the python script `code_VBAP_2D.py` which implements VBAP in 2D. Fill-in required parts (COMPLETE CODE). The script should:

1. Initialize the panner class with a set of 5 loudspeakers corresponding to a 5.1 layout.
2. The `calculate_2d_vbap_gains` should return a list of individual gains for each one of the given loudspeakers layout
3. Angles for virtual sources matching a loudspeaker position (whithin a given "threshold angle for single speaker") will use just one loudspeaker.
4. If two loudspeakers are required, use VBAP in 2D.

To test the function:

1. As panning angle, test all angles from -180° to 180° in intervals of 1° . Print the results.
2. (optional) Plot the values of the coefficients g_i as a function of the panning angle. Plot all the coefficients in the same graph by using five different colors. Check that at most 2 loudspeakers are active for a given angle.
3. Save the finished script, a .txt with the output of the prints of 1), and the plots (if created), in a .zip file and upload it to the aulaglobal Lab2 folder.